North South University

Department of Electrical & Computer Engineering

Fall 2025

Course Title: Analog Electronics I Lab

Faculty Name: Ms. Syeda Sarita Hassan [SSH1]

Section: 7

Experiment Number: 03

Experiment Name:

Zener Diode Applications

Experiment Date: 28-10-2025

Date of Submission: 04-11-2025

Group Number:01

Submitted To: Md. Mahbub Hassan

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Objective :

(i) To study the characteristics of zener diode.

(ii) To observe the diodes behavior in forward, leakage and breakdown region.

(iii) To evaluate how the zener diode maintains a constant output voltage despite changes in input voltage and load resistance.

Theory:

Zener Diode : A heavily doped p-n junction diode that works in both forward bias condition and reverse bias condition is called zener diode. It means, it allows the current to flow in both forward and reverse direction. It is different from normal diodes as normal diodes doesn't work in reverse bias condition. Normal diodes get damaged after reaching the breakdown voltage. But zener diode doesn't get damaged in reverse bias condition. Instead, the voltage across the zener diode become constant after increasing the input voltage to a certain level. Thus, zener diodes are used as voltage regulator.

Fig: zener diode

Load Regulation: Load regulation refers to the ability of a voltage regulator circuit to maintain a constant output voltage even when the load current or load resistance changes. This works because the total current from the source splits between the zener diode and the load. When the load draws more current due to lower resistance, the zener diode draws less current and when the load draws less current, the zener diode draws more current. Thus the voltage across the load resistance remain constant.

Line regulation: Line regulation refers to the ability of a voltage regulator circuit to maintain constant output voltage when the input voltage changes. It is calculated as the change in output voltage divided by the change in input voltage ($\Delta V_{out} / \Delta V_{in}$). A zener diode is used as a voltage regulator by operating in its reverse breakdown region, where it maintains a nearly constant voltage across itself, even when the input voltage fluctuates. So using a load parallel to the zener diode, we can maintain a constant voltage across the load, regardless of the input voltage. But the current flow through the zener diode is limited.

List of Equipment:

Serial No	Component Details	Specification	Quantity
1.	Zener diode	5 volts	1 Piece
2.	Resistor	220Ω, 470Ω, 1kΩ	1 piece each
3.	POT	10kΩ	1 unit
4.	Trainer Board		1 unit
5.	DC Power Supply		1 unit
6.	Digital Multimeter		1 unit
7.	Chords and wire		as required

Circuit Diagram:

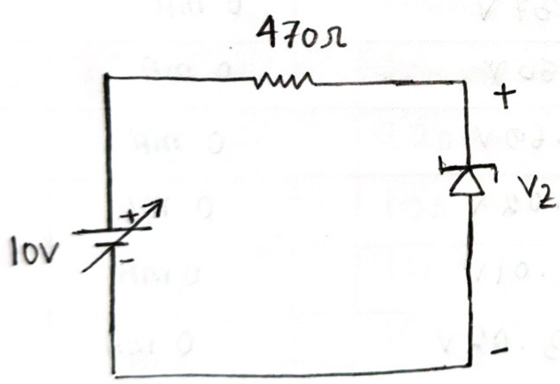

Figure 3.3: Reverse Biased Zener characteristics

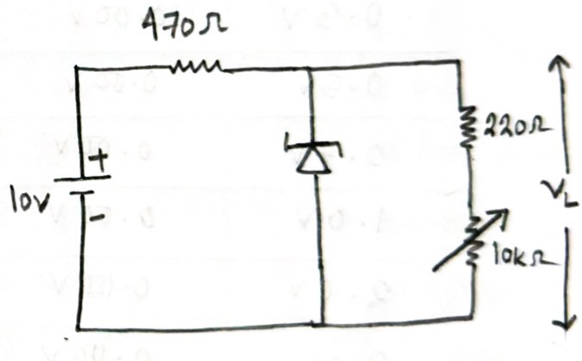

Figure 3.4: Load Regulation

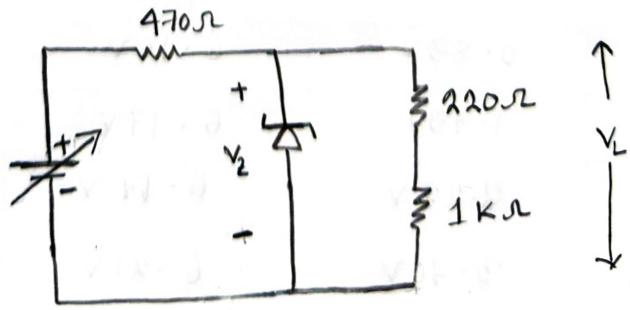

Figure 3.5 : Line Regulation

Data table:

Resistors:

Theoretical	Practical
470Ω	0.461 kΩ / 461Ω
220Ω	0.214 kΩ / 214Ω
1 kΩ	0.977 kΩ / 977Ω

Table 3.1: Data for I-V Characteristics.

V (volts)	V_R (volts)	V_Z (volts)	$I_Z = V_R / R$ (ma)
0.1V	0.00V	0.11V	0 mA
0.3V	0.00V	0.37V	0 mA
0.5V	0.00V	0.50V	0 mA
0.7V	0.00V	0.69V	0 mA
1.0V	0.00V	1.02V	0 mA
2.0V	0.00V	2.01V	0 mA
3.0V	0.00V	3.02V	0 mA
4.0V	0.00V	3.98V	0 mA
5.0V	0.00V	5.03V	0 mA
6.0V	0.04V	5.98V	0.09 mA
7.0V	0.89V	6.16V	1.93 mA
8.0V	1.79V	6.17V	3.88 mA
9.0V	2.22V	6.17V	4.82 mA
10.0V	3.70V	6.21V	8.03 mA

Calculations:

when, $V = 0.5V$, $I_Z = \frac{V_R}{R} = \frac{0V}{0.461\,k\Omega} = 0\,mA$

when, $V = 1V$, $I_Z = \frac{V_R}{R} = \frac{0V}{0.461\,k\Omega} = 0\,mA$

when, $V = 6V$, $I_Z = \frac{V_R}{R} = \frac{0.04V}{0.461\,k\Omega} = 0.09\,mA$

when, $V = 10V$, $I_Z = \frac{V_R}{R} = \frac{3.70V}{0.461\,k\Omega} = 8.03\,mA$

Table 3.2: Data for Load Regulation.

POT R ($k\Omega$)	V_{220} (mV)	V_L (Volts)	I_L (mA) $= V_{220}/220$
1	1140 mV	6.21 V	5.18 mA
3	420 mV	6.21 V	1.91 mA
5	260 mV	6.21 V	1.82 mA
6	220 mV	6.21 V	1.00 mA
7	190 mV	6.21 V	0.86 mA
8	160 mV	6.21 V	0.73 mA
9	150 mV	6.21 V	0.68 mA
10	130 mV	6.21 V	0.59 mA

Table 3.3 : Data for Line Regulation

V (volts)	V_L (volts)
1.0 V	0.77 V
3.0 V	2.21 V
6.0 V	4.36 V
8.0 V	5.76 V
9.0 V	6.17 V
10.0 V	6.17 V
11.0 V	6.19 V
12.0 V	6.21 V

Q:01 Table 3.1: I_z vs V_z graph

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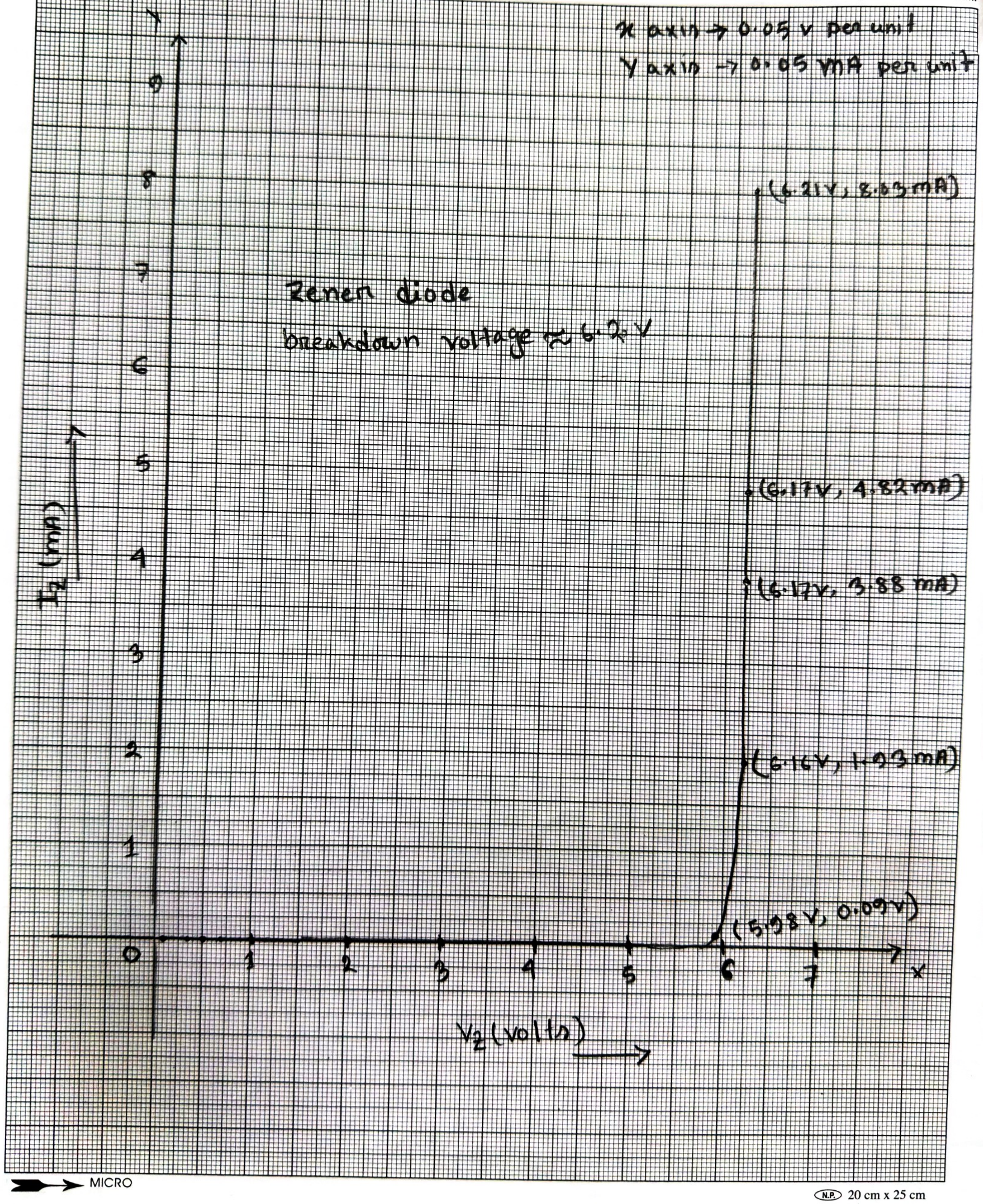

MICRO

N.P. 20 cm x 25 cm

Q2: Table 3.2 : I_L vs V_L

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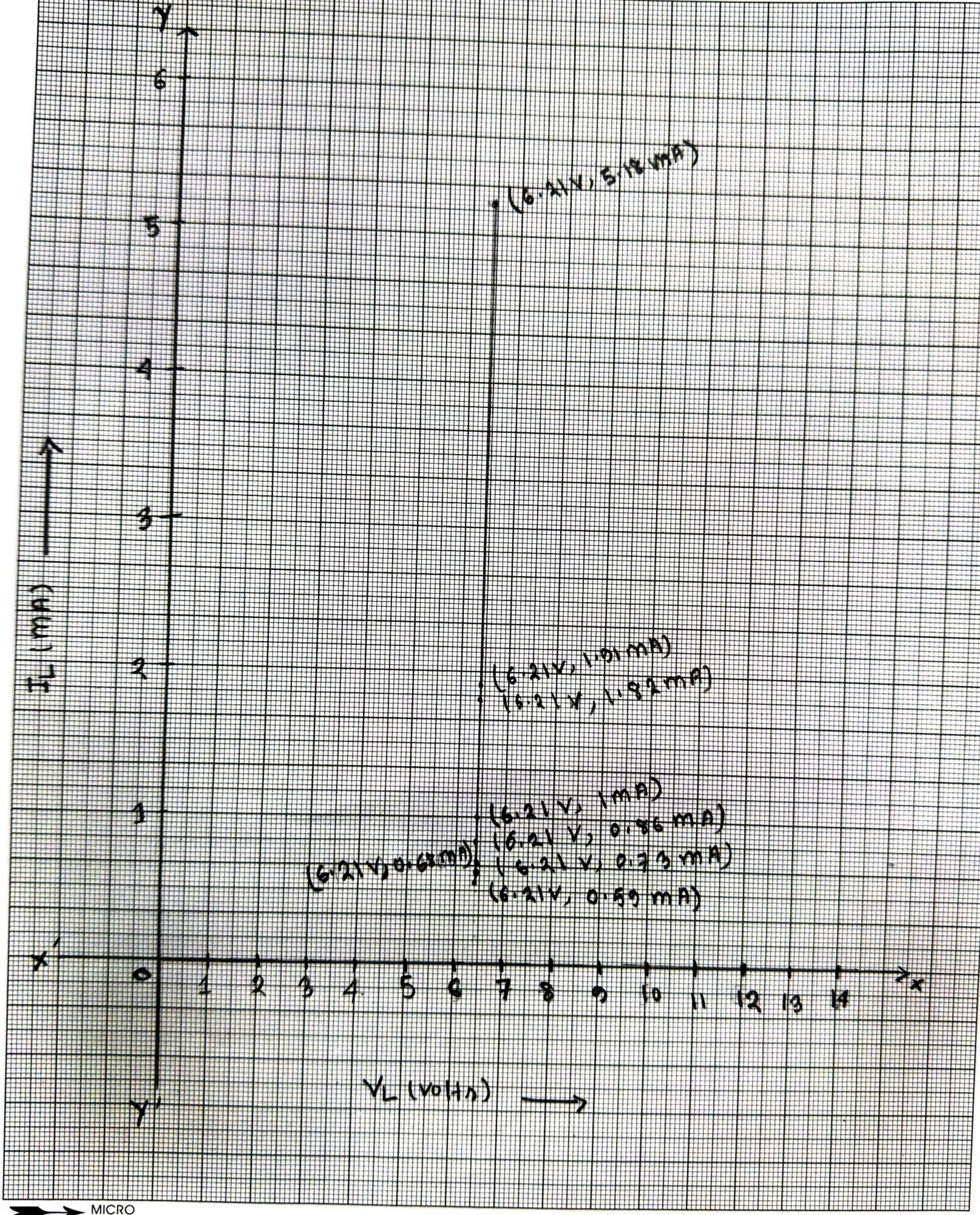
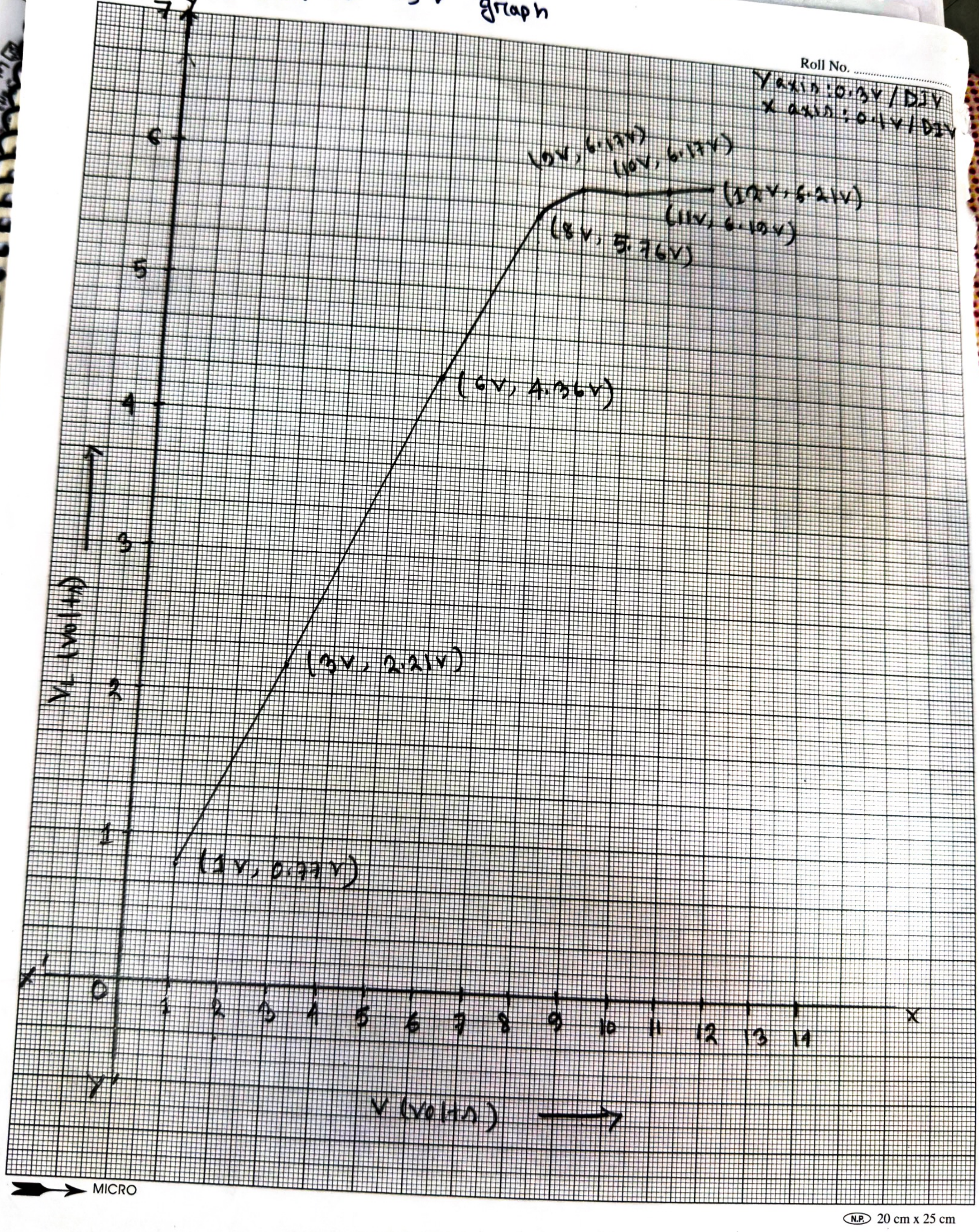


MICRO

N.P. 20 cm x 25 cm

Q3: Table 3.3: V_L vs V graph



Questions and Answers:

(1) From the I_Z vs V_Z graph of table 3.1, we can see that the voltage accross the zener diode remain unchanged after a certain point. This voltage is known as breakdown voltage. According to circuit 01, the breakdown voltage is approximately 6.2 volt.

(2) From the I_L vs V_L graph of table 3.2, we got the load regulation graph. Here we can see that the voltage across the load remain constant and the current across the load changes as the voltage across the resistor 220Ω changes. The graph is vertical.

(3) From the V_L vs V graph of table 3.3, we got the line regulation graph. From the graph, we can see that the value of the voltage across zener diode changes when we keep on increasing the input voltage. But when the Zener diode reaches the zener breakdown voltage, it become constant and doesn't change due to the increase of input voltage. Here Line regulation $= \frac{\Delta V_L}{\Delta V_{in}} = \frac{6.21V - 6.19V}{12V - 11V} = 0.02$

→ If we increase input voltage by 1V then the voltage across the Zener diode increases by 0.02 (almost constant)

Discussion:

From this experiment, we learned how zener diode works in reverse bias direction and how it works for load regulation and line regulation. Here, we observed that the voltage across the zener diode remain constant after the breakdown voltage. Even when we increases the input voltage or resistor the voltage across the zener diode remain constant. During the experiment, we built circuit for both variable load resister and variable input voltage, here we observed that, for variable load resister voltage across the resister remain constant but the across 220Ω resistor changes to make VL constant. For line regulation, the input voltages changes, and the voltage across the 220Ω and 1kΩ resistor changes to keep VL constant after the breakdown voltage.

During the experiment, we faced difficulties using the POT resistor. However, the reading was stable throughout the experiment.

In conclusion, the experiment successfully demonstrated the theorical concept of zener diode and it's applications.

Procedure:

1. Connect the circuit as shown in the figure 3.3
2. Vary the supply voltage from zero volt, complete the Table 3.1.
3. Connect the circuit as shown in the figure 3.4
4. Keep the POT at maximum position and power up the circuit.
5. Gradually decrease the POT resistance and complete the Table 3.2.
6. Replace the POT with 1KΩ resistance, vary the supply voltage and take reading for Table 3.3.

Data Collection:

Signature of instructor:

28-10-2025

Experiment: 3,
Performed by Group# 1

Theoretical	Practical
470Ω	0.461 KΩ / 461 Ω
220Ω	0.214 KΩ / 214 Ω
1KΩ	0.977 KΩ / 977 Ω

Table 3.1: Data for I - V characteristics.

V (volts)	V_R (volts)	V_Z (volts)	$I_Z = V_R / R$ (ma)
0.1	0.00	0.11	0.00
0.3	0.00	0.37	0.00
0.5	0.00	0.50	0.00
0.7	0.00	0.69	0.00
1.0	0.00	1.02	0.00
2.0	0.00	2.01	0.00
3.0	0.00	3.02	0.00
4.0	0.00	3.98	0.00
5.0	0.00	5.03	0.00
6.0	0.04	5.98	0.09
7.0	0.89	6.16	1.93
8.0	1.79	6.17	3.88
9.0	2.22	6.17	4.82
10.0	3.70	6.21	8.03

Table 3.2: Data for Load Regulation

POT R (kΩ)	V220 (mV)	V_L (volts)	I_L (~~Amp~~ mA)= V220/220
1	1140	6.21	5.18
3	420	6.21	1.91
5	260	6.21	1.82
6	220	6.21	1.00
7	190	6.21	0.86
8	160	6.21	0.73
9	150	6.21	0.68
10	130	6.21	0.59

Table 3.3: Data for Line Regulation.

V (volts)	V_L (volts)
1.0	0.77
3.0	2.21
6.0	4.36
8.0	5.76
9.0	6.17
10.0	6.17
11.0	6.19
12.0	6.21

Report:

1. Plot I_Z vs V_Z characteristics of Zener diode. Determine the Zener breakdown voltage from the plot.
2. Plot I_L vs V_L for the data table 4.2. Scale [x-axis: 0.1V/DIV, y-axis: any suitable range]. Find the Load regulation from the graph.
3. Plot V_L vs V for the data table 4.3. Find the line regulation from graph.